Cluster Robust Variance Estimation

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1 Abstract

This thesis studies inference in linear regression models with clustered data, with a particular focus on the consistency and interpretation of cluster-robust variance estimators (CRVE). The

motivation stems from the widespread use of clustering adjustments in applied work and from recent contributions questioning their robustness when clustering is not required by the data-generating process. The first part of the thesis introduces the problem of variance estimation for the OLS estimator and clarifies why clustering matters for asymptotic inference. After defining what clustered data are, the literature review discusses the framework proposed by A. Abadie, S. Athey, G. W. Imbens, and J. M. Wooldridge [1], which reframes clustering as a feature of sampling and assignment design rather than residual correlation, and highlights the conditions under which CRVE or White's heteroskedasticity-robust estimator is appropriate. The core of the thesis develops a general setup with heterogeneous treatment effects across clusters and decomposed error terms. Different assumptions on the number of clusters are considered, together with the role of cluster-level endogeneity. A simulation framework is then introduced to generate both exogenous and endogenous clustered data, allowing population-level exogeneity to hold while within-cluster endogeneity is present. Using Monte Carlo experiments, the thesis studies the consistency of cluster-specific estimators and of the pooled OLS estimator for the average partial effect. The results show that consistency of the average effect can be achieved even in the presence of cluster-level endogeneity, provided that population-level exogeneity holds and all clusters are observed. The final part of the thesis focuses on variance estimation. It analyzes how cluster-level endogeneity affects the asymptotic variance of the pooled estimator and shows why this effect does not average out across clusters. The CRVE matrix is derived formally, its convergence properties are discussed, and its difference from White's variance estimator is made explicit. Monte Carlo evidence illustrates the conditions under which each variance estimator is consistent and clarifies why CRVE should not be used as a default robustness check.

2 Introduction

In costructing the confidence intervals of the OLS estimator $\hat{\beta}_{\text{ols}}$, it is the asymptotic distribution of

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i u_i$$

that determines the estimator's asymptotic distribution. Assuming that $E[X_i u_i] = 0$, the Lindernberg-Levy CLT[2] apply:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i u_i \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{N} \sum_{i} \text{Var }(X_i u_i)\right)$$

But this hold true only if the individual observations are all independent, in fact summing 2 random variables X_1 and X_2 we get another random variable (call it Z) with expected value equal to $E[X_1] + E[X_2]$ and variance equal to $\text{Var }(X_1) + \text{Var }(X_2) + 2 \cdot \text{cov }(X_1, X_2)$. Summing N random variables $(X_1, ..., X_n)$, assume $E[X_i = 0]$, we get a new random variable $Z_N = \sum_{i=1}^{N} X_i$:

$$Z_N \sim \left(0, \sum_{i=1}^{N} \text{Var }(X_i) + 2 \sum_{i \neq}^{N} \sum_{j}^{N} \text{cov }(X_i, X_j)\right)$$

if all the X_is are **independent** all the joint moments (and so also the convariancers) are equal to zero and, dividing by $\sqrt{N}$:

$$\frac{1}{\sqrt{N}}(Z_N) \sim \left(0, \frac{1}{N}\sum_{i=1}^{N} \text{Var }(X_i)\right)$$

So the variance of Z_N scaled by $\frac{1}{\sqrt{N}}$ is the average variance of all the X_is. But here comes the problem, if our data are clustered we do not have that all the observations are independent, but rather we have that the observation of all the individuals inside a cluster are correlated, so we would have that

$$\frac{1}{\sqrt{N}}\sum_{i=1}^{N} X_i u_i \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{N}\sum_{i=1}^{N} \text{Var }(X_i u_i) + \frac{2}{N}\sum_{i\neq}^{N}\sum_{j}^{N} \text{cov }\left(X_i u_i, X_j u_j\right)\right)$$

The variance estimator that is most commonly used, the White variance estimator assumes that all the covariances in the expression above are 0, and so estimates only $\frac{1}{N}\sum_{i=1}^{N} \text{Var }(X_i u_i)$, so it would not consistently estimate the asymptotic variance.

2.1 Why we estimate $\text{Var }\left(\hat{\beta}\right)$

To follow what comes next, it is essential to clarify **what we are actually estimating** when we estimate the variance of $\hat{\beta}_{\text{ols}}$. After all, $\hat{\beta}_{\text{ols}}$ is just a number—so what does it mean to speak of its variance?

Is it a random variable? Does it have a distribution? Yes. The distribution we refer to is the distribution of $\hat{\beta}_{\text{ols}}$ **across repeated samples**, i.e., the values it would take if we changed the individuals drawn from the population. Since $\hat{\beta}_{\text{ols}}$ is a function of random variables (X_i and u_i), if we knew their full population distributions, we would know the exact value of β. But because we only observe a sample, our estimate varies from sample to sample. In this sense, $\hat{\beta}_{\text{ols}}$ *is* a random variable and therefore has a distribution.

2.2 What are clustered data?

Let's take a step back and ask ourselves what does it mean that our data are clustered.

The defining feature of a cluster is that individuals within it exhibit some form of **unobserved** correlation.

Suppose our sample contains N individuals, each with an observed trait X. We can represent these as $X_{ig} = \left(X_{11}, ..., X_{N_1 1}, X_{12}, ..., X_{N_2 2}, ..., X_{N_G G}\right)$, where i indexes individuals within a cluster and g indexes the cluster.

We can think of all these X_{ig} as being drawn from the same population distribution, so they are **identically distributed**. Let us assume this distribution has mean 0.

However, individuals belonging to the same cluster are **not independent**. Their joint distribution is not simply the product of their marginals, their covariances may not be zero.

Saying that observations within a cluster are correlated means that knowing the value of one individual gives information about another individual in the same cluster. If individuals i and j belong to the same cluster g, then:

$$P\big(X_i > E[X] \mid X_j > E[X], i, j \in g\big) > P(X_i > E[X])$$

and similarly:

$$P\big(X_i < E[X] \mid X_j < E[X], i, j \in g\big) > P(X_i < E[X]).$$

outcomes inside the same cluster tend to move together relative to the overall mean—they are positively associated. If we focus on a single cluster g, we can view its observations as:

$$X_g = \{X_i \mid \text{ individual } \; i \in \text{ cluster } \; g\}$$

Each cluster is a subpopulation with its own distribution, and the overall population is the mixture of all G cluster distributions.

3 Literature Review

The common perspective on clustering is what A. Abadie, S. Athey, G. W. Imbens, and J. M. Wooldridge [1] calls the design-based perspective: clustering adjustments were introduced because of correlation in the unobservables. A comprehensive summary of this view is provided by A. C. Cameron and D. L. Miller [3], who offer a definitive treatment of the design-based framework. They emphasize that the key econometric issue is the within-cluster correlation of regression errors: when errors are correlated within clusters but independent across them, conventional OLS standard errors become biased. Their guide clarifies when this bias is particularly acute—namely when regressors are highly correlated within clusters, when error correlation is strong, and when clusters contain many observations. Cluster-robust variance estimators correct this bias by aggregating information *at the cluster level*, and valid inference relies on the number of clusters (not observations) growing. The estimator they describe was first introduced by H. White [4], K.-Y. Liang and S. L. Zeger [5] and is based on large G asymptotics ($G \to \infty$), its asymptotic properties are discussed in details by B. E. Hansen and S. Lee [6]. We will give a breakdown of it in section Section 10.

In settings with a fixed number of clusters G, a useful reference is R. Ibragimov and U. K. Müller [7]. Their main result shows that when the parameter of interest varies across clusters, and when one can consistently estimate each cluster-specific effect, inference can be based on the finite-sample robustness of the classical t-test. If the G cluster-level estimates are (asymptotically) independent and approximately Gaussian—with arbitrarily heterogeneous variances —then a standard t-test on these G estimates remains conservative at conventional significance levels. Under these conditions, hypotheses about the mean of the cluster-specific effects, which is the average partial effect, can be tested using a t_{G-1} distribution, without requiring consistent estimation of the full variance–covariance structure.

The starting point of this thesis is the framework proposed by A. Abadie, S. Athey, G. W. Imbens, and J. M. Wooldridge [1], which reframes clustering adjustments as a problem of sampling and assignment design rather than residual correlation.

The authors argue that the commonly used CRVE is not robust in the following sense: when clustering is not required, the CRVE does not consistently estimate the asymptotic variance of $\hat{\beta}_{\text{OLS}}$, so that using it as a precautionary default is incorrect. Moreover, they emphasize that a discrepancy between the variance estimated using White's heteroskedasticity-robust estimator[8] and that obtained using the CRVE does not, by itself, imply that the CRVE is the correct estimator. So what does?

What the the authors claim is that cluster adjustment matters not because errors or regressors are correlated within clusters, but because of how the data are generated. What drives the need for clustering adjustment is the design: either the sampling scheme, i.e. the probability of including an individual in the sample depends on the cluster she belongs to. Or the experimental design, i.e. treatment assignment itself depends on cluster membership, as in the case of region-specific policies.

In this sense, correlations of residuals or regressors inside clusters are neither necessary nor sufficient to justify cluster-robust adjustments.

They conclude that the choice between correcting for clustering and not must not depende on the observed correlation, but rather on the assumptions we are making about the data generating process.

If neither of the sampling scheme or the assignment mechanism are clustered, then you shall not cluster.

Moreover they claim that cluster robust variace estimator (CRVE from now on) is too conservative unless:

- There are homogeneous effect (i.e. the effect of the treatment is equal for all people in a cluster)
- You only observe a small fraction of clusters
- You only sample at most one individual per cluster (in this case the computed S.E. wuold be equals to the one obtained using White's heteroskedasticity robust variance estimator)

4 General setup

In this section, we lay out the general setup that will guide the rest of the thesis. We will assume the following model

$$y_{ig} = \boldsymbol{\beta}_g \mathbf{X}_{ig} + \gamma \mathbf{Z}_g + \boldsymbol{u}_{ig}$$

So there are two groups of regressors, the one that varies only at the cluster level and the one that varies also at the individual level.

We assume that there are heterogeneous effect across clusters, this means that the true parameter β is different in every cluster.

The average across clusters of β_g we will cal β_{APE}, where APE stands for Average Partial Effect.

We model u_{ig} has having a cluster component and an individual-and-cluster component:

$$u_{ig} = c_g + v_{ig}$$

Where $v_{ig} \sim (0, \sigma_v^2)$, $c_g \sim (0, \sigma_c^2)$ but of course c_g varies only at the cluster level. In this way the expected value of the error given that individual i is in cluster g is:

$$\mathbb{E}_{\mathbb{g}}(u_{ig}) = c_g$$

This can be thought of as a cluster specific intercept, and as long as we include an intercept in our estimated model that varies for each cluster we can assume w.l.o.g that $\mathbb{E}_{\mathrm{g}}\left[u_{ig}\right] = 0 \forall g$.

4.1 Assumptions we make

We must begin by defining precisely the assumptions that we are willing make about the data generating process.

The first choice is whether we treat the number of clusters as fixed and fully observed in every sample—for example, U.S. states (G = 50) when estimating the effect of a federal policy. Alternatively, we may assume an increasing number of clusters, where the specific clusters included can vary from sample to sample.

In both cases there is a two stage sampling scheme going on: in the first stage the clusters are sampled, in the second one the individuals inside the sampled clusters are drawn.

4.1.1 Fixed amount of clusters

- in the first stage all the population parameters are drawn. This include γ the vector of length G with all the β_g, and the cluster specific distributions for X_g and u_g. This stage only happens one, all the parameters from this stage are fixed.
- The second stage is a draw from the cluster distributions of (X, u) of the individual observations that will make our sample. This stage re-happens every time we take a sample from the population, this is the (only) source of randomness that there is in this setup.

4.1.2 Infinite amount of clusters

If the amount of clusters that we possibly have is not finite the first sampling stage reoccurs every time we sample. So there are two sources of randomness:

1. which clusters we draw from the possibly infinite population of clusters and,
2. which individuals we draw from those clusters.

4.2 Endogeneity

Then we must decide whether the sample we are working with presents cluster-level endogeneity, i.e. if

$$\mathbb{E}\left[X_{ig} c_g \mid i \in g\right] \neq 0.$$

In order for $\hat{\beta}_{\text{ols}}$ to be consistent we need exogeneity, meaning that regressors must be uncorrelated with the errors. However, this condition may hold on average (across clusters) while failing within individual clusters: some clusters may display correlation between the cluster-specific component of the error and the regressors, as discussed by Y. Mundlak [9]. When this happens, we may still assume that these correlations can cancel out across clusters, so that exogeneity holds only in the aggregate and a pooled OLS estimator is consistent.

5 Simulating a dataset

We will simulate data in two ways, in the first one we will construct them so that the expected value of the score is 0 in all clusters, in the second one the expected value of the score will be 0 only at the population level.

First we draw a random vector $\boldsymbol{\beta_g}$ from a normal distribution with mean β_{APE} and variance σ^2_β.

```
include("../src/ClusterSimulation.jl")
G = 100 #Number of clusters
Ng = 500 # Number of individuals per cluster

β = rand(Normal(2,4),G)
```

```
100-element Vector{Float64}:
  8.3671018265962
 -3.044037392210061
 -2.4039046955032752
  6.791878246254022
 -0.3510585211341257
 -1.389519282473012
  4.306254686597303
 -3.4977914674642747
 -1.4991562634570492
  2.0431650484451
 -0.44516593234567026
  0.41804290275704337
 -6.537353092813138
  ⋮
 -0.6725082380217433
  4.402591281234272
  1.3517060887903534
 11.220239502583636
  4.653300556833905
 11.77469379190646
 -1.4394696780124243
  2.881933661621998
 -0.6250028727647852
  3.7437212248990166
  2.2210423337084477
  4.2273048772331165
```

We then simulate our dataset in two steps, according to the following model:

$$y_{ig} = \beta_g x_{ig} + \left(c_g + v_{ig}\right)$$

In the first step we simulate the population parameters, and in the second we draw observations from them.

5.1 Description of the Two Steps

```
pop = hePopulation(G, β, randn(G))
```

```
Population with 100 clusters and with heterogeneous effect
```

This line creates a heterogeneous-effects population.
- **`G`** is the number of clusters.
- **`β`** is the vector of betas.
- **`randn(G)`** produces the vector $c_g \sim \mathcal{N}(0, 1)$.

The constructor returns a population struct that stores the cluster-level parameters used to simulate data.

```
sam = sample(pop, Ng)
```

```
Cluster sample struct
```

The line above generates a sample from the population.
- **`Ng`** is the number of observations per cluster.

The function draws data for all `G` clusters, each containing `Ng` observations, and returns a sample struct built according to the population's parameters.
I can then estimate $\hat{\beta}_{\mathrm{g}}$ in each cluster and then broadcast it to all clusters in our sample, getting G estimates.

5.2 Endogenous clusters

The second way in which we'll simulate our data is as follow.
We want to simulate data such that:

$$\mathbb{E}\left[X_{ig} u_{ig} \mid i \in \text{cluster} \quad g\right] = \mu_g$$

for each cluster g, and

$$\mathbb{E}\left[X_{ig} u_{ig}\right] = \mathbb{E}\left[\ \mathbb{E}\left[X_{ig} u_{ig} \mid i \in \text{cluster} \quad g\right]\ \right] = \mathbb{E}\left[\mu_g\right] = 0$$

Across all clusters, the overall expectation is zero because μ_g has mean zero. We first draw from a normal distribution $\mathcal{N}\left(0, \sigma_\mu^2\right)$ a vector $\boldsymbol{\mu}$ of length G, with all the cluster's mean μ_g. Note that in doing so we are taking a sample of size G from a theoretically infinite population of clusters.
Then we:

1. Draw two i.i.d. standard normal variables:

$$Z_1 \sim \mathcal{N}(0, 1), \quad Z_2 \sim \mathcal{N}(0, 1)$$

   Z_1 and Z_2 are zero-mean and independent. The covariance between X and u will come entirely from the shared Z_1 component.
2. Construct within each cluster g:

$$X_g = s_g \, Z_1$$

$$u_g = a_g \, Z_1 + b_g \, Z_2$$

a_g controls the strength of the correlation within the cluster, b_g let u have positive variance.

$$a_g = \frac{\mu_g}{s_g}, \qquad b_g = \sqrt{s_g^2 - a_g^2}.$$

To make sure that b_g exist and is positive we need $s_g^4 \geq \mu_g^2$. So we choose:

$$s_g^2 = |\mu_g| + v_0 \quad \text{with} \quad v_0 > 0$$

So that:

$$\mathbb{E}[X \mid g] = \mathbb{E}[u \mid g] = 0$$

and

$$\mathbb{E}[Xu \mid g] = \text{Cov }(X, u \mid g) = s_g\, a_g = \mu_g.$$

We then compute Y_{ig} choosing a value for β.

$$Y_{ig} = X_{ig}\beta + u_{ig}$$

Let us now check that the expected value of the score is 0 in the whole population but not in each clusters.

```
beta2 = 2
μg = rand(Normal(0,4),500)
endoPop = clPopulation(500, μg, beta2)
endoSam = sample(endoPop,100)
S = checkScore(endoSam)
```

```
500-element Vector{Float64}:
 -1.2362056534515466
 -1.9296425602925564
  2.573662891388222
  1.2544375249160253
 -1.038467104570013
 -2.513045927263646
  0.4987471077735426
  6.245231821129631
  1.0052156005554123
 -2.3358302237888022
  2.1607692708169983
  0.4880018235514524
  0.22744418280513087
  ⋮
  1.267185073777798
 -2.6778590550736703
  0.9786995336736649
 -1.207044653960579
```

```
 1.0691037197304258
 1.410678907080132
-3.644637765053824
 2.9992499693327757
 0.7683613984378522
-1.3584845511944257
-0.1185780260481775
 1.6244325143559708
```

The above is the score in each cluster, now let's check te mean and variance.

```
mean(S), var(S)
```

```
(-0.09615331899955974, 4.7450793580856425)
```

We can see that the mean is close to 0 and the variance is close to the expected variance of μ_g.

6 Consistency

We can discuss two way in which our estimator can be consistent, the first one is that:

$$\widehat{\boldsymbol{\beta}_{\boldsymbol{g}}} \xrightarrow{p} \boldsymbol{\beta}_{\boldsymbol{g}}$$

meaning that each of the G estimated parameters is consistent for its counterpart in the population. This would require that the expected value of the score is 0 in all clusters, i.e. no cluster-level endogeneity. The second way in which our estimator may be consistent is that:

$$\hat{\beta}_{\text{pols}} \xrightarrow{p} \beta_{\text{APE}}$$

For this to be the case we only need population exogeneity.

6.1 Montecarlo without endogeneity

We will now use a Montecarlo method to test wether we can get a consistent estimate of the G parameters if there is no cluster endogeneity. We will estimate the following model:

$$\mathbf{y} = \mathbf{X}_g \hat{\boldsymbol{\beta}}_{Ng} + \mathbf{u}_g$$

Where $\hat{\boldsymbol{\beta}}_{\boldsymbol{g}} = \begin{bmatrix} \alpha_g \\ \beta_g \end{bmatrix}$. Adding a cluster-specific intercept lets the model absorb cluster-level fixed effects in the residuals.

```
histogram(
    permutedims(stack(bols.(sam.allclusters)))[:,2] - β,
    bins = 15
    )
```

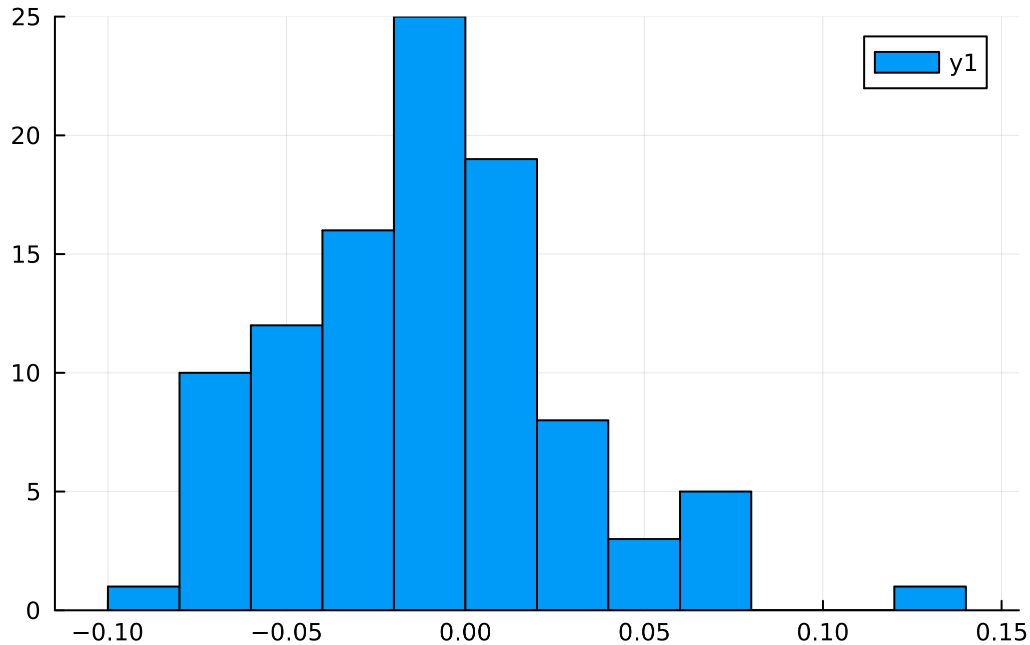


Figure 1: HIstogram of the difference between the estimated parameters and the true one in all clusters

In Figure 1 we can see a plot of the difference between the estimated value for β_g and the true value, it is clear from the graph that all the estimated parameters are consistent for their population counterpart. Now we'll run a Montecarlo simulation, with 100 iterations, we will simulate for each iteration 50 clusters of 100 people and compute $\hat{\beta}_g$ for each cluster. We will then compute the mean across simulation of the difference between the estimated β_g and the true paramenter.

```
betahatDiff = montecarloClusterWise(pop,100,100) .- β
histogram(mean(betahatDiff, dims = 2), bins =15)
```

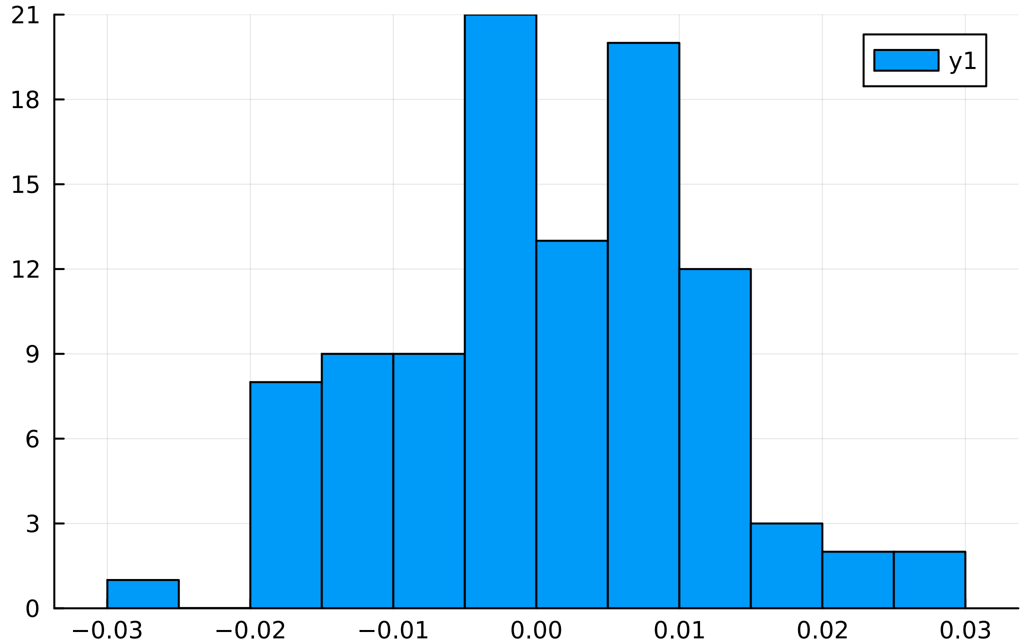


Figure 2: Histogram of the mean difference between estimated and true parameters across simulation in the montecarlo experiment

As can be clearly seen from Figure 2 there is consistency across all clusters. This holds true even if each time we resample the clusters (i.e. the number of clusters in our population is possibly infinite).

6.2 Average Partial Effect

Since β_{APE} is simply the average of all the β_g, if each β_g can be consistently estimated, then β_{APE} can also be consistently estimated. But what happens if there is cluster-level endogeneity?
In that case, consistency is still achievable—as long as endogeneity does not arise at the population level. The key point is that estimating $\hat{\beta}_{\text{pols}}$ is equivalent to averaging the $\hat{\beta}_g$ across clusters. For each cluster we have

$$\widehat{\beta}_g \xrightarrow{p} \beta_g + \frac{\sum_{i \in g} S_i}{\sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i}}$$

where $S_i = X_i u_i$ is the score. Averaging this over all G clusters gives:

$$\hat{\beta}_{\text{pols}} = \frac{1}{G} \sum_g \widehat{\beta}_g \xrightarrow{p} \beta_{\text{APE}} + \frac{\sum_g \sum_{i \in g} S_i}{\sum_g \sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i}}$$

As long as we observe all clusters, the numerator of the second term is equal to 0, and the estimator is therefore consistent.
But if we don't observe all G clusters than we cannot get a consisten estimation of $\widehat{beta}_{\text{pols}}$.

6.3 Consistency as $N_g \xrightarrow{p} \infty$

We want to test wether our estimate of the average partial effect gets better if the number of people that we draw in each cluster increases. In order to do so we ran a series of montecarlo simulation increasing N_g.

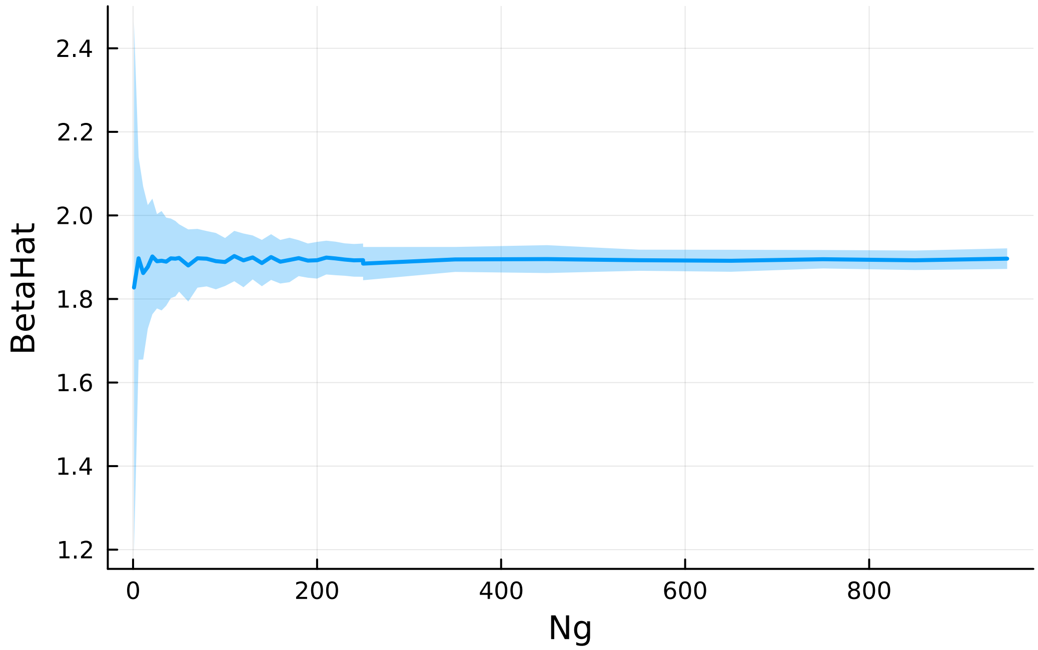


Figure 3: Consistency of betahat as Ng

What we see in Figure 3 is the result of the montecarlo simulation, the line is the average estimates for each N_g, while the shaded area is the standard deviation, we can clearly see from the graph that as the number of observations inside a cluster grows $\hat{\beta}_{\text{ols}} \xrightarrow{p} \beta_{\text{APE}}$.

7 Variance Estimation

In each cluster we have that

$$\widehat{\beta_g} \xrightarrow{p} \beta_g + \frac{\sum_{i \in g} S_i}{\sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i}}$$

Where S_i is the score $X_i u_i$.

$$\left(\widehat{\beta_g} - \beta_g\right) \xrightarrow{d} \mathcal{N}(0, V_g)$$

$$V_g = \sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i} \left(\text{Var} \left(\sum_{i \in g} S_i \right) \right) \sum_{i \in g} \mathbf{x_{i'}} \mathbf{x_i}$$

Since inside each cluster observations can be considered i.i.d.

$$\text{Var}\left(\sum_{i\in g} S_i\right) = \sum_{i\in g}\text{Var}\ (Si)$$

The variance is then

$$\mathbb{E}\left(S_{ig}^2\right) - \left[\mathbb{E}\left(S_{ig}\right)^2\right]$$

Now if we want to estimate β_{APE} we note that it is equal to the average across G of the various β_g, and since the vaious $\widehat{\beta}_g$ are independent, the variance of their mean will be the mean of their variances.

$$\frac{1}{\sqrt{G}}\sum_g\left(\widehat{\beta}_g - \beta_g\right) = \hat{\beta}_{\text{pols}} - \beta_{\text{APE}} \overset{a}{\sim} \mathcal{N}\left(0, \frac{1}{G}\sum_g \text{Var}\ \left(\widehat{\beta}_g\right)\right)$$

As noted before if we have enough clusters:

$$\sum_g\sum_{i\in g} S_i \xrightarrow{p} \mathbb{E}_g\left(S_{ig}\right) = 0$$

So we get a consistent estimate even if there is cluster level endogeneity. But this not apply to the variance. That is because

$$\sum_g\sum_{i\in g} S_i^2 \xrightarrow{p} \mathbb{E}_g\left(S_{ig}\right) \neq 0$$

So that there is not averaging out across clusters that make the second term disappear. This means that the asymptotic variance of $\hat{\beta}_{\text{pols}}$ will be different depending on wether there is cluster-level endogeneity or not, and the difference will be equals to:

$$V_{\text{endo}} - V_{\text{exo}} = \frac{\mathbb{E}_g\left(\mu_g^2\right)}{\sum_i \mathbf{x_{i'}}\mathbf{x_i}}$$

8 Montecarlo

There is a distribution with mean β_{APE} and variance σ_β^2 from which we sample G β_g, obtaining a vector $\boldsymbol{\beta_g}$.

```
G = 50
β = Normal(2,4)
beta1 = rand(β,G)
```

```
50-element Vector{Float64}:
  9.527708346215404
 -1.4291936327623218
  4.66376811712397
```

```
 0.7215674592857257
-0.9734978928124782
 2.310154044001728
-2.3035058184125923
-1.93233669836208
-0.5543940663925886
 6.2986182601095715
12.725972713589648
 4.138791886154299
-0.4642866478179677
 ⋮
 2.3093173504275137
 4.352831366812917
 0.9641474579282912
 3.1199708657147927
 6.301825494138851
 1.5720576536570199
-1.981861355951985
 2.7606613284210586
 2.5882640238733705
 5.598262150992644
 0.29728375987184075
-2.20171171497167
```

$$E[\boldsymbol{\beta_g}] = \begin{bmatrix} E[\beta_1] \\ E[\beta_2] \\ \dots \\ E[\beta_G] \\ \end{bmatrix} = \begin{bmatrix} \beta_{\text{APE}} \\ \beta_{\text{APE}} \\ \dots \\ \beta_{\text{APE}} \end{bmatrix}$$

Assuming that the various β_g are independent we have that across repeated samples:

$$\text{Cov }(\boldsymbol{\beta_g}) = \begin{bmatrix} \sigma_\beta^2 & 0 & \dots & 0 \\ 0 & \sigma_\beta^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \sigma_\beta^2 \end{bmatrix}$$

Taking the mean of the sampled β_g we have that ss $G \to \infty$:

$$\bar{\beta_g} = \frac{1}{G} \sum_{g=1}^{G} \beta_g \xrightarrow{p} \beta_{\text{APE}}$$

$$\frac{1}{\sqrt{G}} \sum_{g=1}^{G} (\beta_g - \beta_{\text{APE}}) \xrightarrow{d} \mathcal{N}\left(0, \sigma_\beta^2\right)$$

If we could observe all the various β_g we could get an estimate of σ_β^2 by doing:

$$\frac{G}{G-1}\sum_{g=1}^{G}\left(\beta_g-\bar{\beta_g}\right)^2$$

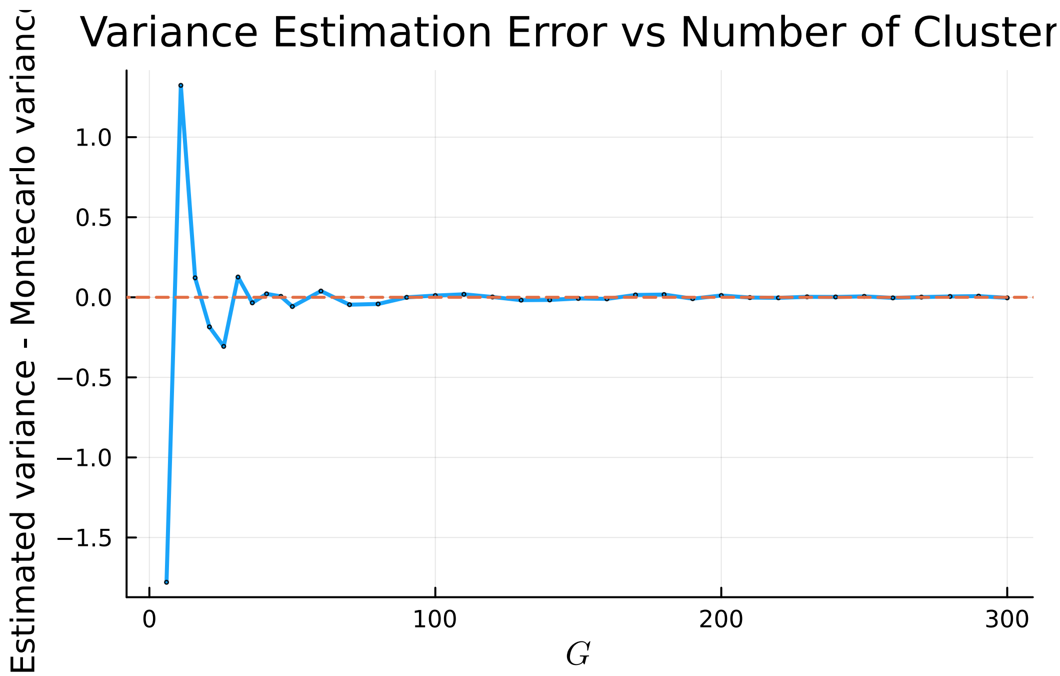
We will now show these things using a Montecarlo method:

```
varianceOfBeta = []
for _ in 1:1000
    beta = rand(β,G)
    barBeta = mean(beta)
    push!(varianceOfBeta,barBeta)
end

estimatedVar = (1 / (G - 1)) * sum((beta1 .- mean(beta1)).^2)/G
estimatedStDev = sqrt(estimatedVar)
print("The estimated standard deviation is: " , std(varianceOfBeta), "\nThe
montecarlo standard deviation is: ", estimatedStDev)
```

```
The estimated standard deviation is: 0.5714417783003277
The montecarlo standard deviation is: 0.5598341622619644
```

```
v = vcat(collect(1:5:50), collect(50:10:300))
l = length(v)

betaDict = Dict()
for G in v
    varianceBeta = []
    for _ in 1:1000
        beta = rand(β,G)
        barbeta = mean(beta)
        push!(varianceBeta, barbeta)
    end
    beta = rand(β,G)
    estimatedVar = (1 / (G - 1)) * sum((beta .- mean(beta)).^2)/G
    var(varianceBeta)
    betaDict[G] = estimatedVar - var(varianceBeta)
end


Gs = sort(collect(keys(betaDict)))
vals = [betaDict[G] for G in Gs]

plot(
    Gs,
    vals,
    seriestype = :line,
    marker = :circle,
```

```
    linewidth = 2,
    markersize = 1,
    alpha = 0.9,
    xlabel = L"G",
    ylabel = "Estimated variance - Montecarlo variance",
    title = "Variance Estimation Error vs Number of Clusters",
    legend = false,
    grid = :on
)

hline!([0.0], linestyle = :dash, linewidth = 1.5)
```



Of course we do not observe all the β_g but we need to estimate them.
Assuming that we can consistently estimate all of the various β_g that we sampled, i.e. that there is no cluster level endogeneity we would obtain an estimated vector $\widehat{\boldsymbol{\beta_g}}$. Each of the element of this vector will be distributed, where here the distribution is considering resampling of individuals, as:

$$\begin{bmatrix} \sqrt{N_1}\left(\widehat{\beta_1} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0, ({X_1}'X_1)^{-1} E_1[{X_1}' u_1 {u_1}' X_1] ({X_1}'X_1)^{-1}\right) \\ \sqrt{N_2}\left(\widehat{\beta_2} - \beta_2\right) \xrightarrow{d} \mathcal{N}\left(0, ({X_2}'X_2)^{-1} E_2[{X_2}' u_2 {u_2}' X_2] ({X_2}'X_2)^{-1}\right) \\ \ldots \\ \sqrt{N_G}\left(\widehat{\beta_G} - \beta_g\right) \xrightarrow{d} \mathcal{N}\left(0, ({X_G}'X_G)^{-1} E_G[{X_G}' u_G {u_G}' X_G] ({X_G}'X_G)^{-1}\right) \end{bmatrix}$$

Where the subscripts index the cluster.
Thus, across clusters, we obtain G asymptotically normal estimators, each centered at its own

true parameter β_g, with cluster-specific asymptotic variances. Each estimated coefficient can be written as

$$\hat{\beta}_g = \beta_g + e_g,$$

where e_g is a mean-zero estimation error satisfying

$$\sqrt{N_g}\, e_g \xrightarrow{d} \mathcal{N}(0, V_g)$$

Hence, the observed object is the vector $\hat{\boldsymbol{\beta}}$, which can be interpreted as noisy realizations of the underlying cluster-level parameters with noise arising from within-cluster sampling variation. Now we show this mechanism with a cluster-wise Monte Carlo experiment.

Using the vector of true parameters that we have previously simulated we construct a population with G clusters. For each cluster, we repeatedly resample individuals and re-estimate the corresponding coefficient $\hat{\beta}_g$. This produces, for every cluster g, a Monte Carlo distribution of $\hat{\beta}_g$ that reflects only within-cluster sampling variation.

The resulting histograms display these empirical distributions, while the vertical line marks the true parameter β_g. This visualization makes explicit that each $\hat{\beta}_g$ can be interpreted as the true β_g plus a mean-zero estimation error, and that the dispersion of the estimator differs across clusters according to their sampling variability.

```
iter = 500
Ng = 100
pop = hePopulation(G,beta1,randn(G))
sam = sample(pop,Ng)
M = montecarloClusterWise(pop,iter,Ng) #G rows, iter columns

plots = [
    begin
        p = histogram(
            M[i, :],
            title = "Cluster $i",
            legend = false
        )
        vline!(p, [beta1[i]], color = :red, linewidth = 2)
        p
    end
    for i in 1:size(M, 1)
]

plot(plots..., layout = (10, 5), size = (1600, 1600))
```

And each β_g can be thought of as β_{APE} plus an i.i.d. cluster specific deviation $d_g \sim \mathcal{N}\left(0, \sigma_\beta^2\right)$, so that:

$$\widehat{\beta_g} = \beta_{\text{APE}} + d_g + e_g$$

The mean across clusters of these estimated parameters is the usual pooled OLS estimator, and it would converge to the average partial effect as $G \to \infty$.

$$\hat{\beta}_{\text{pols}} = \beta_{\text{APE}} + \frac{1}{G}\sum_g d_g + \frac{1}{G}\sum_{g=1}^{G} e_g$$

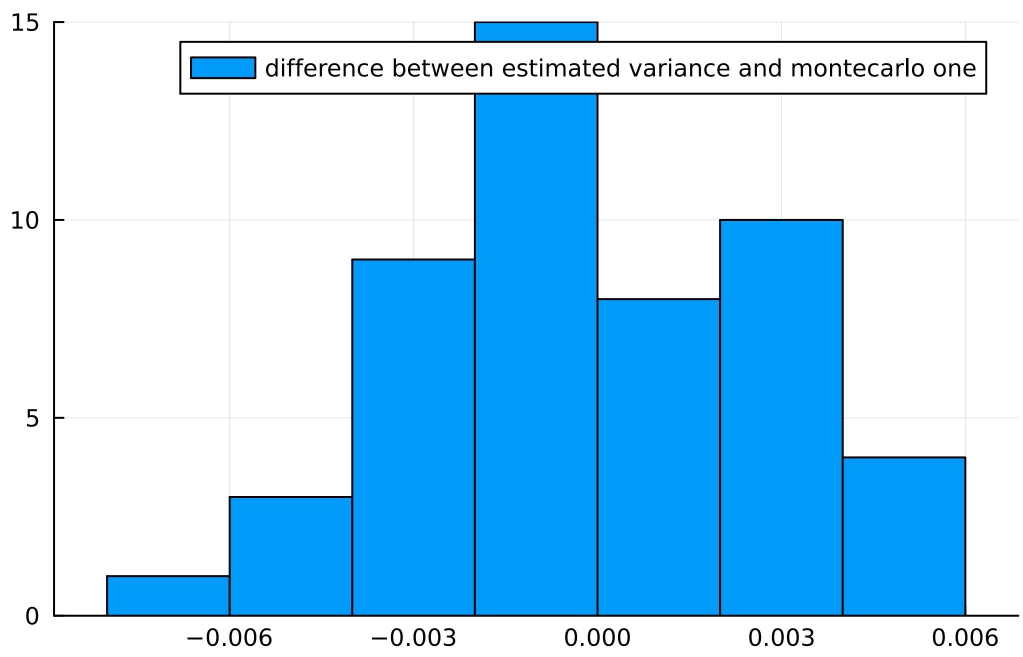
$\frac{1}{G}\sum_{g=1}^{G} d_g \xrightarrow{p} 0, \frac{1}{G}\sum_{g=1}^{G} e_g \xrightarrow{p} 0.$

We now demonstrate, by means of Monte Carlo simulations, that the asymptotic variance of each $\widehat{\beta}_g$ (V_g) can be consistently estimated using the White heteroskedasticity-robust variance estimator.

For each cluster, we compute the White heteroskedasticity-robust variance estimate of $\hat{\beta}_g$. We then compare this quantity to the Monte Carlo variance of $\hat{\beta}_g$, obtained from repeated resampling. The histogram plots the difference between the White-based variance estimate and the Monte Carlo variance across clusters, while the printed average summarizes the mean discrepancy, providing evidence on the consistency of the White estimator for V_g.

```
white = getindex.(WhiteAvar.(sam.allclusters),2,2)
print("The average across clusters difference between the estimated variance
using white and the Montecarlo variance is: ", mean(white - var(M,dims = 2)))
histogram(white .- var(M, dims = 2), label = "difference between estimated
variance and montecarlo one")
```

```
The average across clusters difference between the estimated variance using
white and the Montecarlo variance is: -0.0002449331464512526
```



$$\frac{1}{\sqrt{G}}\sum_{g} d_g \xrightarrow{d} \mathcal{N}\left(0, \sigma_\beta^2\right)$$

$$\sqrt{G}\left(\hat{\beta}_{\text{pols}} - \beta_{\text{APE}}\right) \xrightarrow{d} \mathcal{N}(0,)$$

Note that computing the White variance estimator in the whole population (so for the Asymptotic variance of the pooled OLS) is equivalent to averaging across cluster the estimated variances of each $\widehat{\beta_g}$ using White.

$$\widehat{\mathrm{Var}}\left(\hat{\beta}\right) \approx \frac{1}{G} \times \left(\underbrace{\frac{1}{G} \sum_{g=1}^{G} \widehat{\mathrm{Var}}\left(\hat{\beta}_g\right)}_{\text{Average of Cluster Variances}} \right)$$

Computing the variance of the pooled OLS estimator like this mirrors the E. F. Fama and J. D. MacBeth [10] approach.
But this approach is valid only if there is a one-stage sampling, i.e. if the clusters in my sample are always the same, and so the variance is only given by the estimation error due to within cluster sampling.
We will show this with a montecarlo experiment.
What we aim to show is the following. First, for each cluster-specific estimator $\hat{\beta}_g$, its asymptotic variance can be consistently estimated using the standard heteroskedasticity-robust (White) estimator applied at the cluster level.
Second, we seek to clarify the relationship between applying White's heteroskedasticity-robust estimator to the pooled OLS regression and applying it separately within each cluster. We want to test wether the pooled White variance estimator aggregates information from all clusters in a way that resembles an average of the cluster-specific White estimators.
Finally, we want to demonstrate that when resampling is introduced the variance of the pooled OLS estimator no longer corresponds to the simple average of the variances of the individual estimators $\hat{\beta}_g$ computed via White.

8.1 Decomposing Variation in $\hat{\beta}_g$ Into e_g and d_g

To make the decomposition between within-cluster estimation error e_g and between-cluster heterogeneity d_g explicit, I run two separate simulation exercises.

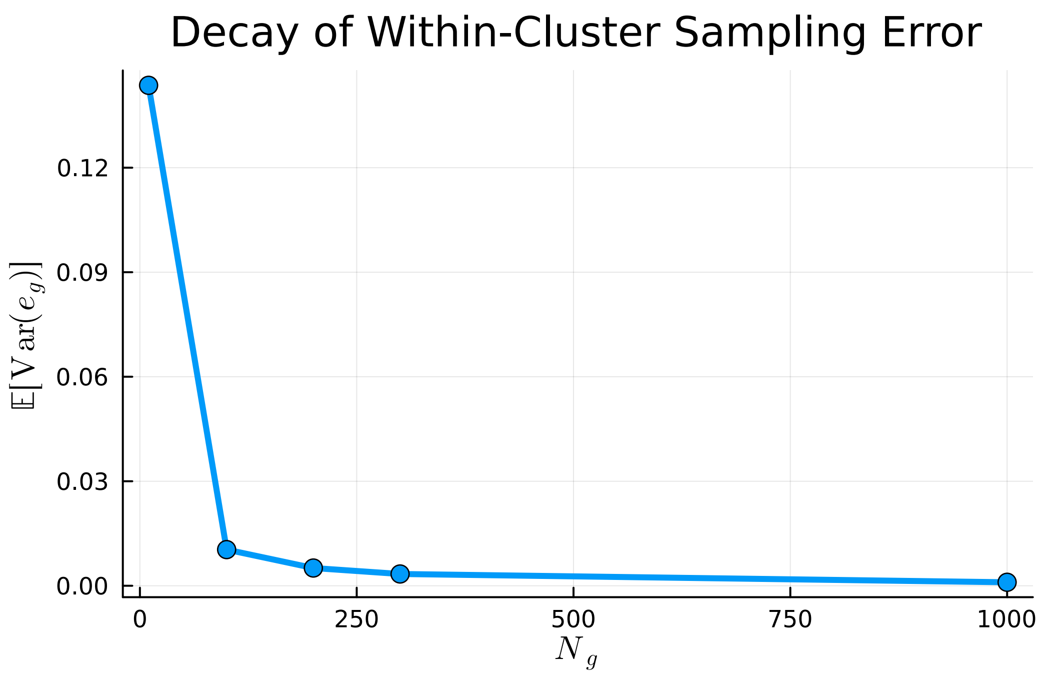
8.1.1 Within-cluster sampling error e_g

In the first exercise, I study how the variance of the estimation error e_g behaves as the cluster size N_g increases.
For a grid of cluster sizes $N_g \in \{10, 100, 200, 300, 1000\}$, I repeatedly resample N_g individuals from the population and re-estimate all cluster-specific coefficients $\hat{\beta}_g$. For each resample, I compute the estimation error

$$e_g = \hat{\beta}_g - \beta_g,$$

and store its empirical variance across clusters. By averaging these variances over many repetitions, I obtain an estimate of $\mathbb{E}\left[\mathrm{Var}\ \left(e_g\right)\right]$ for each N_g.
This simulation shows that the variance of e_g declines rapidly as N_g grows. Larger clusters therefore yield increasingly precise estimates of their own β_g, and the dispersion in $\hat{\beta}_g$ attributable to sampling noise becomes negligible.

```julia
Ngs = [10,100,200,300,1000]
dNg = Dict()
for Ng in Ngs
    d = Dict(i => [] for i in 1:G)
    for _ in 1:1000
        sample1 = sample(pop,Ng)
        eg = getBeta(bols.(sample1.allclusters)) .- beta1
        for g in 1:G
            push!(d[g],eg[g])
        end
    end
    vareg = mean([var(e) for e in values(d)])
    dNg[Ng] = vareg
end
xs = sort(collect(keys(dNg)))
ys = [dNg[n] for n in xs]

plot(
    xs,
    ys;
    xlabel = L"N_g",
    ylabel = L"\mathbb{E}[\mathrm{Var}(e_g)]",
    title = "Decay of Within-Cluster Sampling Error",
    lw = 3,
    ms = 5,
    marker = :circle,
    legend = false,
    grid = :on,
    dpi = 300,
)
```

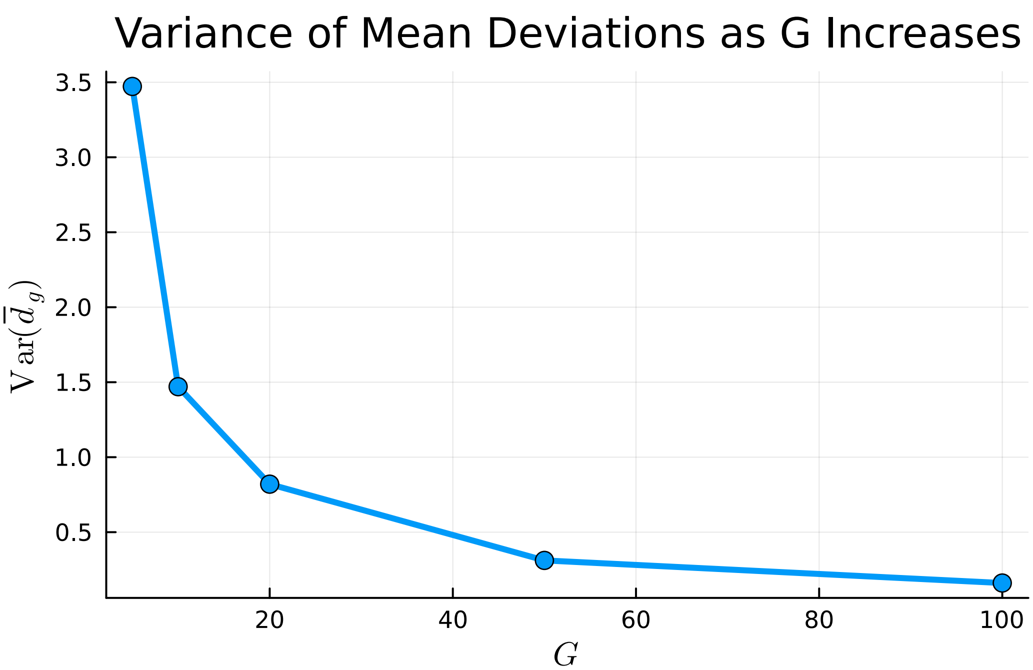


8.1.2 Between-cluster heterogeneity d_g

In the second exercise, I quantify the contribution of the heterogeneity component d_g.
To do this, I repeatedly draw a fresh set of true cluster-level coefficients β_g from the population distribution and compute their deviations from the average partial effect (d_g). For each repetition, I take the mean of these deviations across clusters and record its value.
The sampling distribution of these means provides an estimate of the variance generated by between-cluster differences in the underlying parameters.
This dispersion does not decrease with the cluster size N_g, since it reflects structural heterogeneity rather than sampling variability.
I also replicate this exercise while varying the number of clusters G. For each value of G, I repeatedly draw new sets of heterogeneous coefficients, compute the deviations d_g, and record the variance of their sample mean. This shows how $\mathrm{Var}\left(\overline{d}_g\right)$ decreases as G grows, simply because averaging over more clusters yields a more precise estimate of the underlying mean parameter.

```
Gs = [5, 10, 20, 50, 100]
dG = Dict()

for G in Gs
    mus = Float64[]
    for _ in 1:1000
        beta = rand(β, G)
        dg = beta .- 2
        m = mean(dg)
        push!(mus, m)
    end
```

```
    dG[G] = var(mus)
end

xs = sort(collect(keys(dG)))
ys = [dG[g] for g in xs]

plot(
    xs,
    ys;
    xlabel = L"G",
    ylabel = L"\mathrm{Var}(\overline{d}_g)",
    title = "Variance of Mean Deviations as G Increases",
    lw = 3,
    ms = 5,
    marker = :circle,
    legend = false,
    grid = :on,
    dpi = 300,
)
```



8.2 Decomposition montecarlo

Le us now investigate how well the variance of the pooled OLS estimator can be decomposed into the two components introduced above, the within-cluster estimation error e_g and the between-cluster heterogeneity d_g. To do this, we run a two-dimensional simulation over a grid of cluster sizes $N_g \in \{20, 50, 100, 200, 300\}$ and numbers of clusters $G \in \{5, 10, 20, 50, 100\}$. For each pair (N_g, G), we repeatedly draw a vector of heterogeneous coefficients β_g, generate a synthetic pop-

ulation, and then extract a sample of size N_g from every cluster. In each repetition, we re-estimate all cluster-level coefficients $\hat{\beta}_g$, compute the estimation errors $e_g = \hat{\beta}_g - \beta_g$, the heterogeneity terms $d_g = \beta_g - \beta_{\text{APE}}$, and obtain the pooled OLS estimate $\hat{\beta}_{\text{ols}}$.
We record:

1. the sampling variability of the e_g's,
2. the dispersion of the d_g's,
3. the variance of the pooled estimator across repetitions, and
4. the White and CRVE variance estimates for each sample. With these objects in hand, we directly test the decomposition

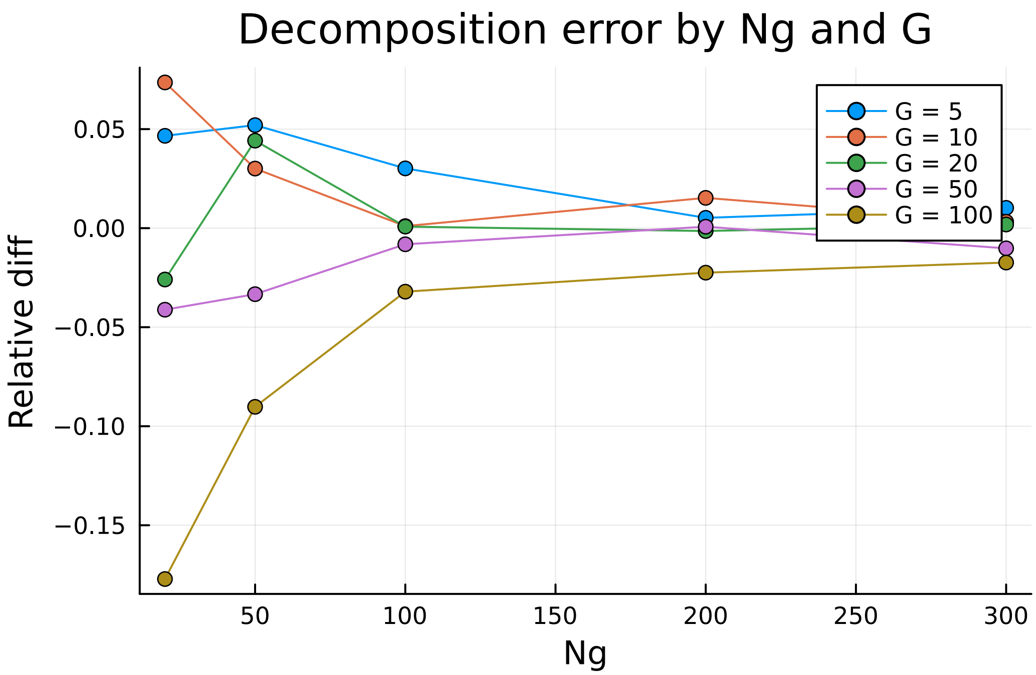
$$\text{Var}\left(\hat{\beta}_{\text{ols}}\right) \approx \text{Var}\left(d_g\right) + \mathbb{E}\left[\text{Var}\left(e_g\right)\right]$$

where the right-hand side is obtained from the simulated cluster-level quantities. For every (N_g, G), we compare the empirical variance of $\hat{\beta}_{\text{ols}}$ with the sum of these two components and compute the relative discrepancy. The results reveal a very clear pattern. When either the number of clusters or the cluster size is large, the decomposition works almost perfectly: both the heterogeneity term d_g and the sampling component e_g are measured very precisely, so their sum closely matches the variability of the pooled estimator. When both G and N_g are small, however, the approximation naturally worsens. In this case, the cluster-level estimates are noisier, the differences between clusters appear more dispersed than they truly are, and the overall variability of $\hat{\beta}_{\text{ols}}$ is not yet captured well by the decomposition. The plot of the relative error across (N_g, G) makes this pattern clear, showing that the mismatch quickly fades as we increase either the number of clusters or their size.

```julia
Ngs = [20, 50, 100, 200, 300]
Gs = [5, 10, 20, 50, 100]

crve_white_diff = Dict()
total_var = Dict()
egDict = Dict()
dgDict = Dict()
betaDict = Dict()

for G in Gs
    for Ng in Ngs

        eg_mat = Float64[]
        dg_mat = Float64[]
        betaHatOls = Float64[]
        crve_est = Float64[]
        white_est = Float64[]

        for _ in 1:1000
            beta1 = rand(β,G)
            pop = hePopulation(G,beta1,randn(G))
```

```
            sam = sample(pop, Ng)
            beta_hat_g = getBeta(bols.(sam.allclusters))

            betaHat = bols(sam)
            push!(betaHatOls, betaHat[2])

            eg = beta_hat_g .- beta1
            push!(eg_mat, var(eg))

            dg = beta1 .- 2
            push!(dg_mat, mean(dg))

            push!(crve_est, CRVE(sam)[2,2])
            push!(white_est, whiteAvar(sam)[2,2])
        end

        diff = (mean(crve_est) - mean(white_est))
        crve_white_diff[(Ng, G)] = diff

        dgDict[(Ng,G)] = dg_mat
        egDict[(Ng,G)] = eg_mat
        betaDict[(Ng,G)] = betaHatOls
        total_var[(Ng, G)] = var(eg_mat) + var(dg_mat)
    end
end

var_decomp = Dict()

for G in Gs
    for Ng in Ngs
        b   = betaDict[(Ng, G)]
        egv = egDict[(Ng, G)]
        dgm = dgDict[(Ng, G)]

        lhs = var(b)
        rhs = var(dgm) + mean(egv)

        var_decomp[(Ng, G)] = (
            var_bols        = lhs,
            var_dg_plus_eg  = rhs,
            diff            = lhs - rhs,
            rel_diff        = (lhs - rhs) / lhs
        )
    end
end

plt = plot(xlabel = "Ng", ylabel = "Relative diff", title = "Decomposition
error by Ng and G",ms= 5)
```

```
for G in sort(Gs)
    rels = [var_decomp[(Ng, G)].rel_diff for Ng in sort(Ngs)]
    plot!(sort(Ngs), rels, marker = :circle, label = "G = $G")
end

display(plt)
```



9 CRVE–White Differences as an Estimator of Between-Cluster Heterogeneity

Finally, we test the hypothesis that the difference between the CRVE and White variance estimators isolates the component generated by between-cluster heterogeneity, namely $\mathrm{Var}(\bar{d}_g)$. For each design (N_g, G), we compute the average CRVE variance, the average White variance, take their difference, and compare it to the simulated value of $\mathrm{Var}(\bar{d}_g)$. The scatterplot displays this comparison, with the horizontal axis reporting the variance of the mean deviations d_g and the vertical axis reporting the difference between the two variance estimators. Each point corresponds to a specific pair (N_g, G), and its color intensity reflects the number of clusters G, with lighter markers indicating larger G.

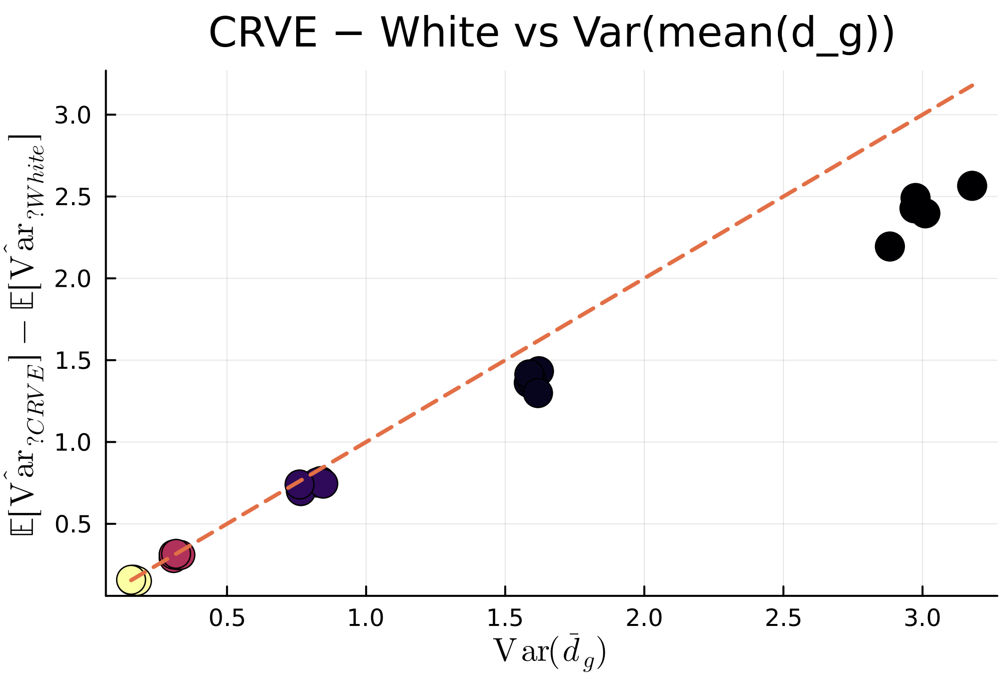
The figure shows a remarkably tight alignment between the two quantities: the points lie very close to the 45-degree dashed line, indicating that the CRVE–White difference almost exactly recovers $\mathrm{Var}(\bar{d}_g)$ across all designs. The pattern is especially strong when G is large—these observations appear as the lighter points, sitting nearly on the identity line—which reflects the fact that averaging across many clusters allows us to estimate the heterogeneity component with high precision. When G is small, the points are slightly more dispersed, but still follow the same linear relationship, confirming that the gap between CRVE and White is driven by between-cluster variation rather than sampling noise. Overall, the simulation provides direct evidence that

$$\mathbb{E}\left[\widehat{\mathrm{Var}}_{\mathrm{CRVE}}\right]\mathbb{E}\left[\widehat{\mathrm{Var}}_{\mathrm{White}}\right] \approx \mathrm{Var}\left(\bar{d}_g\right),$$

and that this relationship strengthens as the number of clusters grows.

```
dgVar = Dict{Tuple{Int,Int},Float64}()

for ((Ng, G), dg_vals) in dgDict
    dgVar[(Ng, G)] = var(dg_vals)
end

# Compare diff (CRVE - White) to Var(mean(dg))
results = Dict{Tuple{Int,Int},NamedTuple}()

for key in keys(crve_white_diff)
    diff = crve_white_diff[key]
    vdg  = dgVar[key]

    results[key] = (
        diff              = diff,
        var_mean_dg       = vdg,
        abs_diff          = diff - vdg,
        rel_error         = (diff - vdg) / vdg,
    )
end

# Extract data
Ng_vals = [k[1] for k in keys(results)]
G_vals  = [k[2] for k in keys(results)]
diffs   = [results[k].diff for k in keys(results)]
vdgs    = [results[k].var_mean_dg for k in keys(results)]

scatter(
    vdgs,
    diffs;
    marker_z = G_vals,       # COLOR based on G
    xlabel = L"\mathrm{Var}(\bar d_g)",
    ylabel = L"\mathbb{E}[\widehat{\mathrm{Var}}_{\text{CRVE}}] - \mathbb{E}[\widehat{\mathrm{Var}}_{\text{White}}]",
    title = "CRVE – White vs Var(mean(d_g))",
    ms = 8,
    colorbar_title = L"G",
    legend = false,
)
```

```
plot!(identity; lw=2, ls=:dash, label=false)
```



9.1 All cluster observed and no endogeneity

To understand how the different variance estimators behave in finite samples, we simulate repeated datasets under four designs and compare the resulting Monte Carlo distribution of $\hat{\beta}_{\text{ols}}$ with the normal approximations implied by White and by CRVE. The procedure is straightforward: for each design we take the empirical estimate from the original sample, then repeatedly redraw data from the corresponding population (with or without resampling), generating many realizations of $\hat{\beta}_{\text{ols}}$. The histogram shows the empirical sampling distribution across these replications, while the two curves plot the normal densities using the White and CRVE standard errors. This allows us to directly assess which estimator better captures the true sampling variability of the coefficient.

```
betahat = bols(sam)[2]
white = sqrt(whiteAvar(sam)[2,2])
crve = sqrt(CRVE(sam)[2,2])

mc_results = montecarlo(pop, 1000, Ng, true, false)

histogram(mc_results, bins=50, normalize=:pdf, label="Montecarlo")

d = Normal(betahat, white)
D = Normal(betahat, crve)

x = range(betahat - 4 * crve, betahat + 4 * crve; length = 1000)
```

```
y = pdf.(d, x)
Y = pdf.(D, x)

plot!(x, y, linewidth = 2, label = "White")
plot!(x, Y, linewidth = 2, label = "CRVE")
```

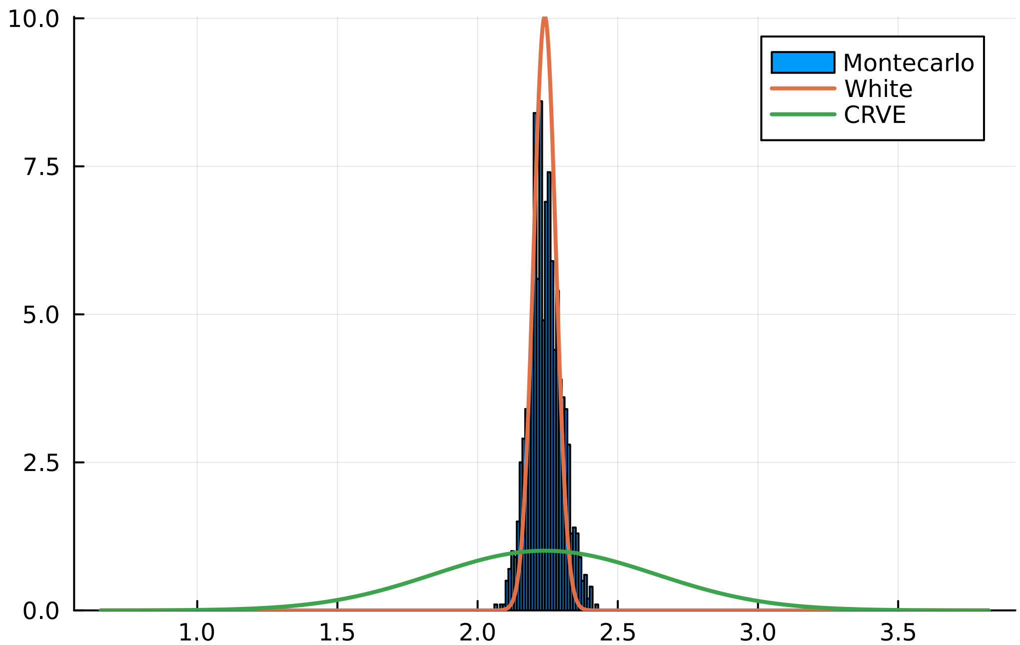
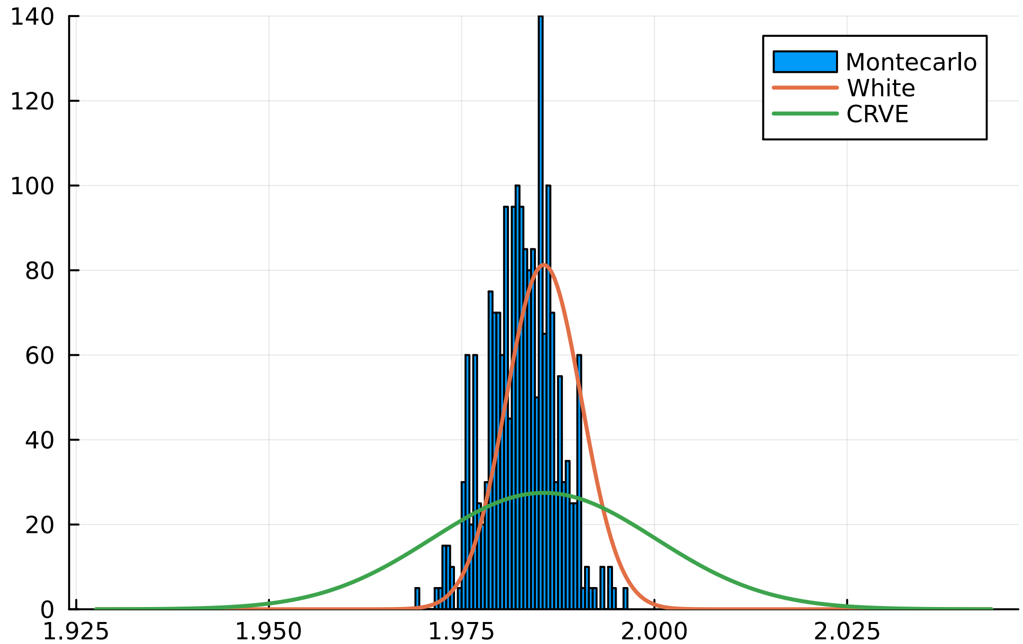


Figure 4: Consistency of White Variance estimator without cluster level endogeneity

In the first design, where there is no cluster-level endogeneity and the sample is not resampled across simulations, the sampling distribution of $\hat{\beta}_{\text{ols}}$ is extremely concentrated around the true parameter. The Monte Carlo histogram forms a tight and symmetric peak, and the White-based normal approximation matches it almost perfectly. In contrast, the CRVE curve is far too wide, overstating the true sampling variability. This confirms the theoretical prediction: when clusters are exogenous and fixed, White provides a consistent estimate of the asymptotic variance of the pooled estimator, while CRVE is unnecessarily conservative.

```
betahat = bols(endoSam)[2]
white = sqrt(whiteAvar(endoSam)[2,2])
crve = sqrt(CRVE(endoSam)[2,2])
histogram(montecarlo(endoPop,400,Ng,true,false), bins=50,normalize=:pdf,
label="Montecarlo")
d = Normal(betahat, white)
D = Normal(betahat, crve)

x = range(betahat - 4 * crve, betahat + 4 * crve; length = 1000)
```

```
y = pdf.(d,x)
Y = pdf.(D,x)
plot!(x,y, linewidth =2, label = "White")
plot!(x,Y, linewidth =2, label = "CRVE")
```

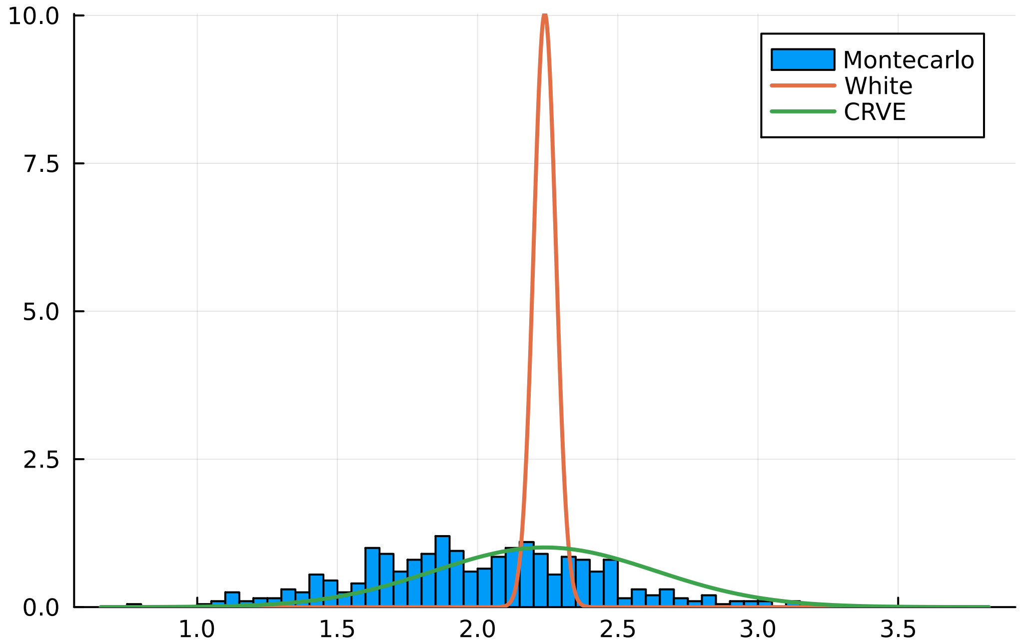


In the second design, with cluster-level endogeneity but no resampling, the Monte Carlo distribution becomes somewhat wider, as expected when the true slopes differ across clusters. Even in this setting, however, White continues to provide the better approximation: its curve closely matches the peak and overall concentration of the histogram. By contrast, the CRVE curve is still far too wide, overstating the sampling variability. Endogeneity introduces extra dispersion, but not enough to overturn the conclusion that White remains the more accurate estimator in this fixed-sample design.

Figure 4

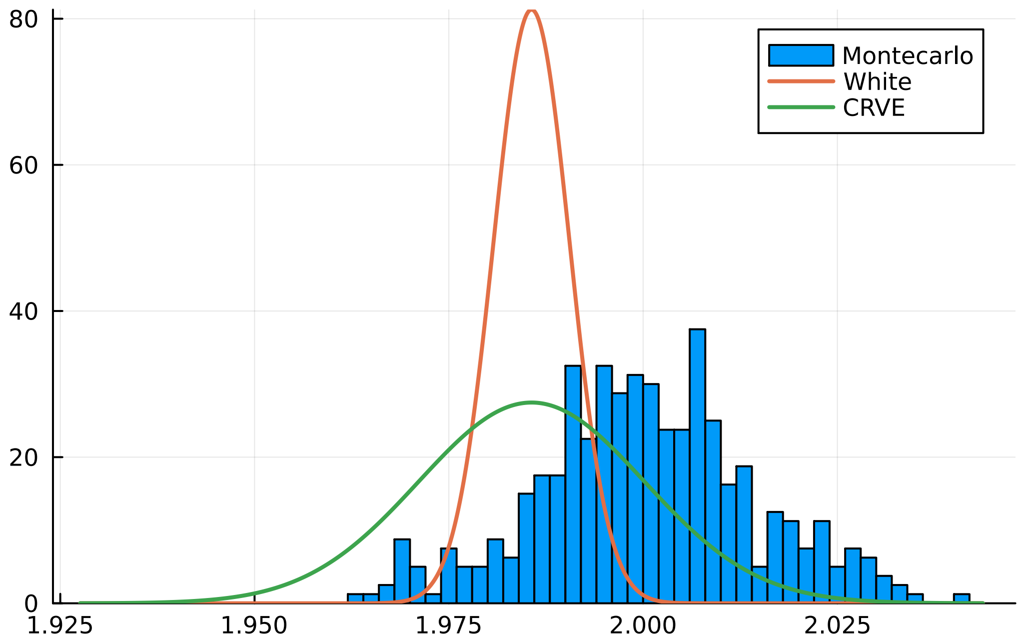
```
betahat = bols(sam)[2]
white = sqrt(whiteAvar(sam)[2,2])
crve = sqrt(CRVE(sam)[2,2])
histogram(montecarlo(pop,400,Ng,true,true),bins=50,normalize=:pdf,
label="Montecarlo")
d = Normal(betahat, white)
D = Normal(betahat, crve)
x = range(betahat - 4 * crve, betahat + 4 * crve; length = 1000)
y = pdf.(d,x)
Y = pdf.(D,x)
```

```
plot!(x,y, linewidth =2, label = "White")
plot!(x,Y, linewidth =2, label = "CRVE")
```



In the third design, there is no endogeneity but clusters are resampled across simulations. This induces much more variability in the Monte Carlo distribution, which becomes wide and highly dispersed. In this setting, White is far too narrow: its curve completely misses the broad shape of the histogram and substantially understates the true sampling uncertainty. By contrast, CRVE aligns much better with the empirical distribution, capturing both the width and the overall spread of the estimates. Once cluster resampling is introduced, CRVE clearly becomes the more appropriate variance estimator.

```
betahat = bols(endoSam)[2]
white = sqrt(whiteAvar(endoSam)[2,2])
crve = sqrt(CRVE(endoSam)[2,2])
histogram(montecarlo(endoPop,400,Ng,true,true), bins=50,normalize=:pdf,
label="Montecarlo")
d = Normal(betahat, white)
D = Normal(betahat, crve)

x = range(betahat - 4 * crve, betahat + 4 * crve; length = 1000)
y = pdf.(d,x)
Y = pdf.(D,x)
plot!(x,y, linewidth =2, label = "White")
plot!(x,Y, linewidth =2, label = "CRVE")
```



In the fourth design, where both cluster-level endogeneity and cluster resampling are present, the Monte Carlo distribution is the widest of all four cases. The combination of heterogeneous slopes and changing cluster composition substantially increases sampling variability. In this environment, White is clearly too narrow and fails to reflect the substantial dispersion seen in the histogram. CRVE, instead, tracks the scale of the distribution much more accurately, providing a noticeably better approximation to the true sampling variation.

10 CRVE Matrix

10.1 Notation

We use clusters indexed by $g = 1, \ldots, G$, and units within clusters indexed by $i = 1, \ldots, N_g$. Each observation has k regressors.

$$\mathbf{X}_{ig} = \left[x_{1ig},\ x_{2ig},\ \ldots\ x_{kig}\right]$$

This is the row vector of regressors for unit i in cluster g.

$$\mathbf{X}_g = \begin{bmatrix} \mathbf{X}_{1g} \\ \mathbf{X}_{2g} \\ \ldots \\ \mathbf{X}_{N_g g} \end{bmatrix} = \begin{bmatrix} x_{1ig} & x_{2ig} & \ldots & x_{kig} \\ x_{1jg} & x_{2jg} & \ldots & x_{kjg} \\ \ldots & & & \\ x_{1N_g g} & x_{2N_g g} & \ldots & x_{kN_g g} \end{bmatrix}$$

This matrix has dimension $N_g \times k$ and stacks all regressors in cluster g. Full regressor matrix stacks all the clusters one on top of the other

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \\ \dots \\ \mathbf{X}_G \end{bmatrix}$$

Equivalently, written by expanding the blocks:

$$\mathbf{X} = \begin{bmatrix} \mathbf{X_{ig}} \\ \mathbf{X_{jg}} \\ \dots \\ \mathbf{X_{N_g g}} \\ \mathbf{X_{ih}} \\ \mathbf{X_{jh}} \\ \dots \\ \mathbf{X_{N_h h}} \\ \dots \\ \mathbf{X_{iG}} \\ \mathbf{X_{jG}} \\ \dots \\ \mathbf{X_{N_G G}} \end{bmatrix}$$

This is an $N \times k$ matrix, where $N = \sum_{g=1}^{G} N_g$, obtained by stacking clusters one after another. The key intuition underlying cluster-robust variance estimation (CRVE) is that dependence is allowed within clusters, while clusters themselves are assumed to be asymptotically independent. Consequently, asymptotic inference is driven by the number of clusters rather than by the total number of individual observations.

This reasoning is consistent with the general principle emphasized by R. Ibragimov and U. K. Müller [7], namely that the consistency of robust variance estimators relies on averaging over an asymptotically large number of essentially uncorrelated units. In the case of clustered standard errors, consistency is achieved by aggregating information across an increasing number of independent clusters, in direct analogy with heteroskedasticity-robust estimators averaging over independent disturbances or long-run variance estimators averaging over weakly dependent frequency components. Assuming that $\mathbb{E}\left[\mathbf{X}_\mathbf{g}'\mathbf{u}_\mathbf{g}\right] = 0$, $\hat{\beta}_{\text{ols}}$ will be a consistent estimator of the population parameter β if $G \to \infty$ [2]. To understand this note that computing β:

$$(X'X)^{-1}(X'y)$$

is the same as doing:

$$\hat{\boldsymbol{\beta}}_{\text{ols}} = \left(\sum_{g}^{G} \mathbf{X_g}'\mathbf{X_g}\right)^{-1} \left(\sum_{g}^{G} \mathbf{X_{g'}y_g}\right)$$

so

$$\left(\hat{\boldsymbol{\beta}}_{\text{ols}} - \boldsymbol{\beta}\right) = \left(\frac{1}{G}\sum_{g}^{G} \mathbf{X_g}'\mathbf{X_g}\right)^{-1} \left(\frac{1}{G}\sum_{g}^{G} \mathbf{X_{g'}}\mathbf{u_g}\right)$$

and applying the Weak Law of Large Numbers J. M. Wooldridge [2]

$$\frac{1}{G}\sum_{g}^{G} \mathbf{X_{g'}}\mathbf{u_g} \xrightarrow{p} \mathbb{E}\left[\mathbf{X}_{\mathbf{g}}'\mathbf{u_g}\right] = 0$$

so $\hat{\boldsymbol{\beta}}_{\text{ols}}$ will be consistent. Now let us derive it's asymptotic distribution:

$$\sqrt{G}\left(\hat{\boldsymbol{\beta}}_{\text{ols}} - \boldsymbol{\beta}\right) = \left(\frac{1}{G}\sum_{g}^{G} \mathbf{X_g}'\mathbf{X_g}\right)^{-1} \left(\frac{1}{\sqrt{G}}\sum_{g}^{G} \mathbf{X_{g'}}\mathbf{u_g}\right)$$

The first term is by assumption $A^{-1} + o_p(1)$, while the second is $\overset{a}{\sim} \mathcal{N}\left(0, \mathbb{E}\left[\mathbf{X_{g'}}\mathbf{u_g}\mathbf{u_{g'}}\mathbf{X_g}\right]\right)$ So

$$\sqrt{G}\left(\hat{\boldsymbol{\beta}}_{\text{ols}} - \boldsymbol{\beta}\right) \xrightarrow{d} \mathcal{N}\left(0, A^{-1}\mathbb{E}\left[\mathbf{X_{g'}}\mathbf{u_g}\mathbf{u_{g'}}\mathbf{X_g}\right]A^{-1}\right)$$

10.2 Formula

$$\hat{V}_{\text{clu}}\hat{\boldsymbol{\upbeta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{B}(\mathbf{X}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{X})^{-1}\left(\sum_{g=1}^{G} \mathbf{X}_g{}'\hat{\mathbf{u}}_g\hat{\mathbf{u}}_g{}'\mathbf{X}_g\right)(\mathbf{X}'\mathbf{X})^{-1}$$

Define:

$$\mathbf{s_g} \equiv \mathbf{X_g}'\hat{\mathbf{u}}_{\mathbf{g}} = \begin{bmatrix} X_{11g} & X_{12g} & \cdots & X_{1N_g g} \\ X_{21g} & X_{22g} & \cdots & X_{2N_g g} \\ \vdots & \vdots & \ddots & \vdots \\ X_{k1g} & X_{k2g} & \cdots & X_{kN_g g} \end{bmatrix} \begin{bmatrix} \hat{u}_{1g} \\ \hat{u}_{2g} \\ \vdots \\ \hat{u}_{N_g g} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{N_g} X_{1ig}\,\hat{u}_{ig} \\ \sum_{i=1}^{N_g} X_{2ig}\,\hat{u}_{ig} \\ \vdots \\ \sum_{i=1}^{N_g} X_{kig}\,\hat{u}_{ig} \end{bmatrix}$$

Then:

$$\mathbf{s_g}\mathbf{s_g}' = \left(X_g{}'u_g\right)\left(X_g{}'u_g\right)' = X_{g'}u_g u_{g'}X_g$$

$$\mathbf{s_g}\mathbf{s_g}' = \begin{bmatrix} \sum_{i\in g}\sum_{j\in g} X_{1ig}X_{1jg}\hat{u}_{ig}\hat{u}_{jg} & \sum_{i\in g}\sum_{j\in g} X_{1ig}X_{2jg}\hat{u}_{ig}\hat{u}_{jg} & \cdots & \sum_{i\in g}\sum_{j\in g} X_{1ig}X_{kjg}\hat{u}_{ig}\hat{u}_{jg} \\ \sum_{i\in g}\sum_{j\in g} X_{2ig}X_{1jg}\hat{u}_{ig}\hat{u}_{jg} & \sum_{i\in g}\sum_{j\in g} X_{2ig}X_{2jg}\hat{u}_{ig}\hat{u}_{jg} & \cdots & \sum_{i\in g}\sum_{j\in g} X_{2ig}X_{kjg}\hat{u}_{ig}\hat{u}_{jg} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i\in g}\sum_{j\in g} X_{kig}X_{1jg}\hat{u}_{ig}\hat{u}_{jg} & \sum_{i\in g}\sum_{j\in g} X_{kig}X_{2jg}\hat{u}_{ig}\hat{u}_{jg} & \cdots & \sum_{i\in g}\sum_{j\in g} X_{kig}X_{kjg}\hat{u}_{ig}\hat{u}_{jg} \end{bmatrix}$$

Then we sum this over all G clusters:

$$\hat{\mathbf{B}} = \begin{bmatrix} \sum_{g=1}^{G} \sum_{i\in g} \sum_{j\in g} X_{1ig}\, X_{1jg}\, \hat{u}_{ig}\, \hat{u}_{jg} & \sum_{g=1}^{G} \sum_{i\in g} \sum_{j\in g} X_{1ig}\, X_{3jg}\, \hat{u}_{ig}\, \hat{u}_{jg} & \cdots & \sum_{g=1}^{G} \sum_{i\in g} \sum_{j\in g} X_{1ig}\, X_{kjg}\, \hat{u}_{ig}\, \hat{u}_{jg} \\ \sum_{g=1}^{G} \sum_{i\in g} \sum_{j\in g} X_{2ig}\, X_{1jg}\, \hat{u}_{ig}\, \hat{u}_{jg} & \sum_{g=1}^{G} \sum_{i\in g} \sum_{j\in g} X_{2ig}\, X_{3jg}\, \hat{u}_{ig}\, \hat{u}_{jg} & \cdots & \sum_{g=1}^{G} \sum_{i\in g} \sum_{j\in g} X_{2ig}\, X_{kjg}\, \hat{u}_{ig}\, \hat{u}_{jg} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{g=1}^{G} \sum_{i\in g} \sum_{j\in g} X_{kig}\, X_{1jg}\, \hat{u}_{ig}\, \hat{u}_{jg} & \sum_{g=1}^{G} \sum_{i\in g} \sum_{j\in g} X_{kig}\, X_{3jg}\, \hat{u}_{ig}\, \hat{u}_{jg} & \cdots & \sum_{g=1}^{G} \sum_{i\in g} \sum_{j\in g} X_{kig}\, X_{kjg}\, \hat{u}_{ig}\, \hat{u}_{jg} \end{bmatrix}$$

So what CRVE does is to compute for each cluster the outer product of $\hat{\mathbf{S}}_{\mathbf{g}} = \mathbf{X}_{\mathbf{g}'} \hat{\mathbf{u}}_{\mathbf{g}}$ with itself, and then average those matrices across clusters.
After pre and post-multiplication by $\left(\frac{1}{N} \sum_i \mathbf{x}_i \mathbf{x}_i' \right)^{-1}$, this converges to the **asymptotic variance-covariance matrix** of the OLS estimator $\hat{\boldsymbol{\beta}}_{\text{ols}}$.

10.3 Convergence

We want to show that

$$\frac{1}{G} \sum_g \mathbf{X}_g'\, \hat{\mathbf{u}}_g \hat{\mathbf{u}}_g' \mathbf{X}_g \xrightarrow{p} \mathbb{E}\left[\mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g\right]$$

First let's substitute for $\hat{\mathbf{u}}_{\mathbf{g}}$:

$$\frac{1}{G} \sum_g \mathbf{X}_g' \left(\mathbf{y}_g - \mathbf{X}_g \hat{\boldsymbol{\beta}} \right) \left(\mathbf{y}_g - \mathbf{X}_g \hat{\boldsymbol{\beta}} \right)' \mathbf{X}_g$$

$$\frac{1}{G} \sum_g \mathbf{X}_g' \left(\mathbf{X}_g \boldsymbol{\beta} + \mathbf{u}_g - \mathbf{X}_g \hat{\boldsymbol{\beta}}_g \right) \left(\mathbf{X}_g \boldsymbol{\beta} + \mathbf{u}_g - \mathbf{X}_g \hat{\boldsymbol{\beta}}_g \right)' \mathbf{X}_g$$

We then factor terms in order to isolate $\left(\beta - \hat{\beta}\right)$

$$\frac{1}{G} \sum_g \mathbf{X}_g' \left[\mathbf{X}_g \left(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_g \right) + \mathbf{u}_g\right] \left[\mathbf{X}_g \left(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_g\right) + \mathbf{u}_g\right]' \mathbf{X}_g$$

As $\hat{\beta} \xrightarrow{p} \beta$, the extra terms vanish, leaving only $\mathbf{u}_g$.

$$\xrightarrow{p} \frac{1}{G} \sum_g \mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g \xrightarrow{p} \mathbb{E}\left[\mathbf{X}_g' \mathbf{u}_g \mathbf{u}_g' \mathbf{X}_g\right]$$

10.4 Difference from White

Starting from the variance–covariance matrix of the estimated score,

$$\hat{S} \hat{S}' = X' \hat{u} \hat{u}' X,$$

whose (i, j)-th element is $x_i x_j\, \hat{u}_i \hat{u}_j$, we can think of White's estimator and the cluster-robust estimator as applying different *selector matrices* to this full matrix of cross-products.

For White, the selector keeps only the diagonal elements. The inner matrix becomes

$$\text{diag}\ \left(\hat{u}_1^2, ..., \hat{u}_n^2\right),$$

so only terms with $i = j$ are retained. Computing White cluster-wise (i.e., within each cluster and then summing) selects exactly the same diagonal elements as doing it pooled.

For the cluster-robust variance estimator (CRVE), the selector keeps all pairs (i, j) such that observations i and j belong to the same cluster. This retains all within-cluster covariances and zeroes all cross-cluster terms.

To see this concretely, consider one cluster with three observations $i = 1, 2, 3$. The **White** inner matrix for that cluster is:

$$B_{\mathrm{W}} = \begin{bmatrix} \hat{u}_1^2 & 0 & 0 \\ 0 & \hat{u}_2^2 & 0 \\ 0 & 0 & \hat{u}_3^2 \end{bmatrix}.$$

The CRVE inner matrix is:

$$B_{\mathrm{C}} = \begin{bmatrix} \hat{u}_1^2 & \hat{u}_1\hat{u}_2 & \hat{u}_1\hat{u}_3 \\ \hat{u}_1\hat{u}_2 & \hat{u}_2^2 & \hat{u}_2\hat{u}_3 \\ \hat{u}_1\hat{u}_3 & \hat{u}_2\hat{u}_3 & \hat{u}_3^2 \end{bmatrix}.$$

After pre- and post-multiplying by the cluster design matrix X_g, the **White** term becomes:

$$X_{g'} B_{\mathrm{W}} X_g = x_1^2\hat{u}_1^2 + x_2^2\hat{u}_2^2 + x_3^2\hat{u}_3^2 = \sum_{i \in g} x_i^2 \, \hat{u}_i^2.$$

For CRVE:

$$\begin{aligned} X_{g'} B_{\mathrm{C}} X_g = {} & x_1^2\hat{u}_1^2 + x_2^2\hat{u}_2^2 + x_3^2\hat{u}_3^2 \\ & + 2(x_1 x_2 \hat{u}_1\hat{u}_2 + x_1 x_3 \hat{u}_1\hat{u}_3 + x_2 x_3 \hat{u}_2\hat{u}_3). \end{aligned}$$

Thus,

$$X_{g'} B_{\mathrm{C}} X_g = X_{g'} B_{\mathrm{W}} X_g + 2 \sum_{\substack{i,j \in g \\ i<j}} x_i x_j \, \hat{u}_i \hat{u}_j.$$

The only difference between White and CRVE is exactly these additional within-cluster cross-products. These terms capture intra-cluster dependence and are the mechanism through which CRVE adjusts for correlated errors within clusters.

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